

SUMMARY OF THE INVENTION

The present disclosure provides systems and methods for generating a user-specific penile anatomical geometry profile through coordinated time-of-flight (ToF) distance ranging and optical motion sensing. In various embodiments, a scanning device is configured to measure radial distance between a sensor reference plane and a penile surface while simultaneously detecting longitudinal displacement along the anatomical axis. By combining spatial distance data with motion-tracked positional information, the system reconstructs a structured three-dimensional geometric representation of the scanned anatomy.

In certain embodiments, one or more ToF sensors are positioned within a housing structure configured to maintain a defined spatial relationship relative to the anatomical surface during measurement. The ToF sensors may operate using direct time-of-flight, indirect time-of-flight, phase-shift measurement, or structured light ranging techniques. Distance samples may be captured at predefined intervals or continuously during motion along the longitudinal axis.

Optical motion sensing components may include optical flow sensors, speckle-based motion detectors, emitter-receiver displacement systems, or other surface-relative tracking technologies capable of detecting controlled movement of the device relative to the anatomy. In some embodiments, inertial measurement units (IMUs) may supplement motion tracking to compensate for angular deviation, tilt, or unintended user movement during scanning.

The system processes collected distance and motion data using computational reconstruction algorithms configured to interpolate intermediate surface points, generate cross-sectional profiles, and assemble a continuous geometric model along the scanned path. Reconstruction techniques may include cylindrical approximation modeling, spline-based interpolation, mesh generation, curvature analysis, and volumetric integration. Error correction routines may be applied to remove motion artifacts or outlier distance samples.

The resulting user-specific anatomical geometry profile may include derived structural metrics such as longitudinal length, radial distribution across multiple cross-sections, taper gradients, curvature mapping, volumetric estimation, symmetry indices, proportionality ratios, and other parametric geometric descriptors. The generated profile may be stored digitally, compared across time to assess structural changes, utilized for device calibration or fitment optimization, incorporated into compatibility modeling systems, or integrated into broader biometric analytics platforms.

In some embodiments, the geometry profiling system may function independently as a standalone measurement platform. In other embodiments, the generated geometry profile may interface with additional physiological sensing systems, comparative modeling engines, or personalization frameworks within a multi-device ecosystem. The disclosure is not limited to any specific housing configuration, sensor arrangement, or reconstruction algorithm, and encompasses variations in form factor, sensing modality, and computational implementation that achieve the described geometry profiling functionality.